

ASSIGNMENT #03: EXPLORATORY DATA ANALYSIS OF ROW CROP FIELDS

ASSIGNMENT OVERVIEW

FIELD	VALUE
ASSIGNMENT NUMBER	3
TITLE	Exploratory Data Analysis of Row Crop Fields
GRADE VALUE	10 points
DUE DATE	March 9, 2026, 11:59 PM PT

OBJECTIVES

After completing this assignment, you will be able to:

- Conduct **Exploratory Data Analysis (EDA)** on your row crop field datasets.
- Use Python (pandas, seaborn, or plotly) to uncover patterns in soil, yield, or weather data.
- Identify correlations, such as how soil pH or organic matter relates to yield potential.
- Translate raw data patterns into visual insights for a farm manager.

INSTRUCTIONS

Step 1: Create Your Feature Branch

Open your terminal in VS Code and switch to a new branch for your analysis:

```
git checkout -b feature/assignment-03-eda
```

Step 2: Set Up Your Analysis Notebook

1. **Create a new notebook** in your repository: `notebooks/03_field_eda.ipynb`.
2. **Load your data:** Import the `field_summary.csv` you created in Assignment #2.
3. **Data Inspection:** Use `.describe()` and `.info()` to check the statistical distributions of your fields.

Step 3: Visual Exploration (The "3 Visuals" Task)

Using `matplotlib`, `seaborn`, or `plotly`, create at least three visualizations that explore the relationships in your data.

- **Visual 1 (Distribution):** Show the distribution of a key variable, such as total acreage or soil organic matter across your 200 fields.

- **Visual 2 (Relationship):** Create a scatter plot exploring the relationship between two variables (e.g., Soil Type vs. Acreage).
- **Visual 3 (Correlation):** Generate a heatmap to see which variables are most closely linked.

Step 4: Design Your Dashboard Elements

The goal of this course is a final dashboard. You must now "lock in" two plots for that final output.

1. Refine two of your plots from Step 3.
2. Add professional titles, axis labels, and a consistent color palette (e.g., agricultural greens or soil browns).
3. **Task:** Save these as high-resolution images in `output/dashboard_assets/`.

Step 5: Update Your Project Tracker

Open `docs/project/assignment-01-tracker.md`.

- Document the key takeaway from your EDA (e.g., *"Fields with Clay-type soils showed 15% higher variability in moisture"*).
- Log any "data cleaning" steps you had to repeat after seeing outliers in your plots.

GRADING CRITERIA (10 POINTS TOTAL)

CRITERION	POINTS	DESCRIPTION
EDA QUALITY	4	Created at least 3 distinct visualizations that reveal meaningful data patterns.

TECHNICAL EXECUTION	2	Clean Python code using standard libraries (Pandas/Seaborn/Plotly).
DASHBOARD DESIGN	2	Designed 2 polished plots specifically for the final dashboard layout.
INTERPRETATION	2	Provided a brief written summary of the relationships found in the data.